

Lab 9

FOOD DELIVERY DATABASE SYSTEM REPORT

Prepared By Raghad Al-bogami

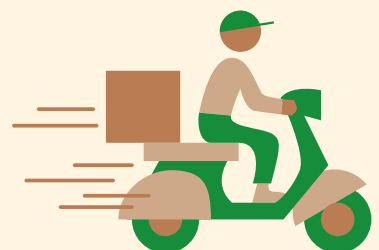


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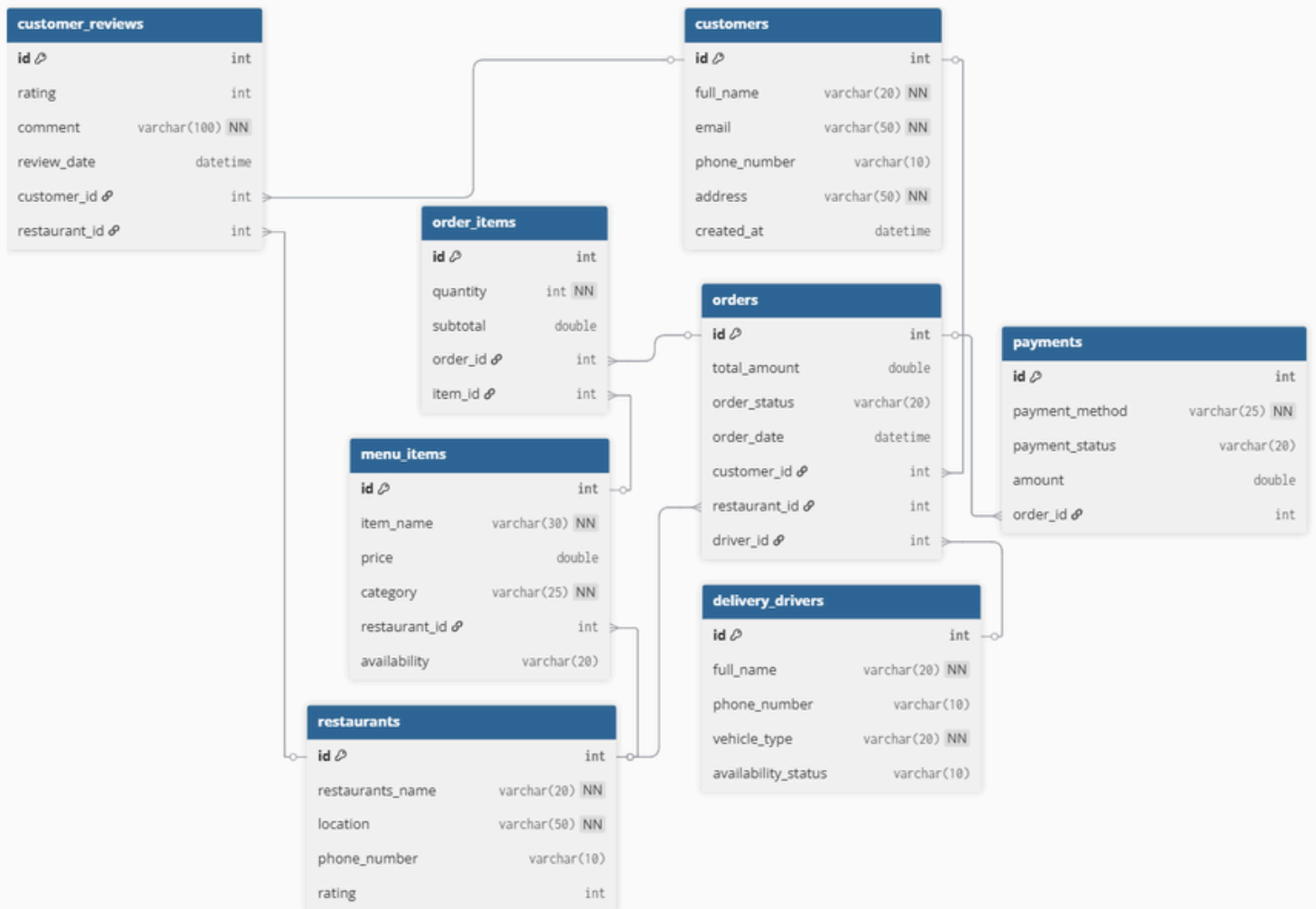
SYSTEM BRIEF

This system is a Food Delivery Database System designed to manage and organize the operations of an online food ordering platform. It includes key entities such as customers, restaurants, menu items, orders, order items, payments, delivery drivers, and customer reviews.

The purpose of the system is to efficiently store and manage data using a relational database model. It ensures data integrity through primary and foreign keys and maintains proper relationships between all tables.

The system allows customers to browse restaurants, place orders, and make payments, while delivery drivers handle the delivery process. This design provides a structured and efficient way to manage food delivery operations.

DATABASE SCHEMA



Note: A clearer and higher-quality version of the schema diagram can be found in the GitHub repository where this report was uploaded.

SCHEMA ANALYSIS

The database consists of several related tables designed to support a Food Delivery System, where all entities are connected using primary and foreign keys to maintain data integrity and proper relationships.

Relationship	Type	Description
customers → orders	One-to-Many (1:M)	One customer can place many orders
restaurants → menu_items	One-to-Many (1:M)	One restaurant has many menu items
restaurants → orders	One-to-Many (1:M)	One restaurant receives many orders
orders → order_items	One-to-Many (1:M)	One order can contain multiple items
menu_items → order_items	One-to-Many (1:M)	One item can appear in many order items
orders → payments	One-to-One (1:1)	Each order has one payment record
customers → reviews	One-to-Many (1:M)	One customer can write many reviews
restaurants → reviews	One-to-Many (1:M)	One restaurant can receive many reviews
delivery_drivers → orders	One-to-Many (1:M)	One driver can deliver many orders

SQL CODE

#create database

```
create database food_delivery;
```

#create tables

```
create table customers(  
    id int primary key ,  
    full_name varchar(20) not null,  
    email varchar(50) unique not null,  
    phone_number varchar(10) unique ,  
    address varchar(50) not null,  
    created_at datetime default current_timestamp  
);
```

```
create table restaurants(  
    id int primary key ,  
    restaurants_name varchar(20) unique not null,  
    location varchar(50) not null,  
    phone_number varchar(10) unique ,  
    rating int check ( rating between 1 and 5)  
);
```

```
create table delivery_drivers(  
    id int primary key ,  
    full_name varchar(20) not null,  
    phone_number varchar(10) unique ,  
    vehicle_type varchar(20) not null,  
    availability_status varchar(10) default 'available'  
);
```

```
create table menu_items(  
    id int primary key ,  
    item_name varchar(30) not null,  
    price double check ( price > 0 ),  
    category varchar(25) not null,  
    restaurant_id int,  
    availability varchar(20) default 'available',  
    foreign key (restaurant_id) references restaurants(id)  
);
```

```
create table orders(  
    id int primary key ,  
    total_amount double check ( total_amount > 0 ),  
    order_status varchar(20) default 'pending',  
    order_date datetime default current_timestamp,  
    customer_id int,  
    restaurant_id int,  
    driver_id int,  
    foreign key (customer_id) references customers(id),  
    foreign key (restaurant_id) references restaurants(id),  
    foreign key (driver_id) references delivery_drivers(id)  
);
```

```
create table order_items(  
    id int primary key ,  
    quantity int not null check ( quantity > 0 ),  
    subtotal double check ( subtotal > 0 ),  
    order_id int ,  
    item_id int,  
    foreign key (order_id) references orders(id),  
    foreign key (item_id) references menu_items(id)  
);
```

```
create table payments(  
    id int primary key ,  
    payment_method varchar(25) not null,  
    payment_status varchar(20) default 'pending',  
    amount double check ( amount > 0 ),  
    order_id int,  
    foreign key (order_id) references orders(id)  
);
```

```
create table reviews(  
    id int primary key ,  
    rating int check ( rating between 1 and 5),  
    comment varchar(100) not null,  
    review_date datetime default current_timestamp,  
    customer_id int,  
    restaurant_id int,  
    foreign key (customer_id) references customers(id),  
    foreign key (restaurant_id) references restaurants(id)  
);
```

#insert tables data

insert into customers values

```
(1,'ahmed ali','ahmed@gmail.com','0500000001','riyadh - al-  
narjis',current_timestamp),  
(2,'sara mohammed','sara@gmail.com','0500000002','jeddah - al-  
safa',current_timestamp),  
(3,'khalid saeed','khalid@gmail.com','0500000003','dammam - al-  
shati',current_timestamp),  
(4,'noor ahmad','noor@gmail.com','0500000004','riyadh - al-  
malqa',current_timestamp),  
(5,'lama khaled','lama@gmail.com','0500000005','mecca - al-  
awali',current_timestamp),  
(6,'fahad nasser','fahad@gmail.com','0500000006','tabuk - al-  
murooj',current_timestamp),  
(7,'reem ali','reem@gmail.com','0500000007','jeddah - al-  
rehab',current_timestamp),  
(8,'omar saad','omar@gmail.com','0500000008','riyadh - al-  
olaya',current_timestamp);
```

insert into restaurants values

```
(1,'burger house','riyadh - king fahd road','0511111111',4),  
(2,'pizza zone','jeddah - tahliya street','0522222222',5),  
(3,'shawarma time','dammam - corniche','0533333333',4),  
(4,'kfc express','riyadh - exit 5','0544444444',5),  
(5,'taco spot','mecca - aziziyah','0555555555',3),  
(6,'noodle bar','jeddah - al-hamra','0566666666',4),  
(7,'grill house','tabuk - al-rashid','0577777777',5),  
(8,'sushi world','riyadh - al-muhammadiyah','0588888888',4);
```

insert into delivery_drivers values

```
(1,'ali mohammed','0502000001','car','available'),  
(2,'saad ahmed','0502000002','bike','available'),  
(3,'nasser ali','0502000003','car','busy'),  
(4,'faisal omar','0502000004','bike','available'),  
(5,'yousef khalid','0502000005','car','available'),  
(6,'mohammed saad','0502000006','bike','busy'),  
(7,'ahmed nasser','0502000007','car','available'),  
(8,'saleh fahad','0502000008','bike','available');
```


insert into menu_items values

```
(1,'burger',25,'fast food',1,'available'),  
(2,'pizza',40,'italian',2,'available'),  
(3,'shawarma',15,'arabic',3,'available'),  
(4,'chicken meal',30,'fast food',4,'available'),  
(5,'taco',20,'mexican',5,'available'),  
(6,'noodles',35,'asian',6,'available'),  
(7,'grill steak',50,'grill',7,'available'),  
(8,'sushi set',60,'japanese',8,'available');
```

insert into orders values

```
(1,50,'pending',current_timestamp,1,1,1),  
(2,80,'delivered',current_timestamp,2,2,2),  
(3,40,'pending',current_timestamp,3,3,3),  
(4,100,'delivered',current_timestamp,4,4,4),  
(5,60,'pending',current_timestamp,5,5,5),  
(6,70,'delivered',current_timestamp,6,6,6),  
(7,90,'pending',current_timestamp,7,7,7),  
(8,120,'delivered',current_timestamp,8,8,8);
```

insert into order_items values

```
(1,2,50,1,1),  
(2,1,40,2,2),  
(3,3,45,3,3),  
(4,2,60,4,4),  
(5,1,20,5,5),  
(6,2,70,6,6),  
(7,1,50,7,7),  
(8,2,120,8,8);
```

insert into payments values

```
(1,'cash','paid',50,1),  
(2,'card','paid',80,2),  
(3,'cash','pending',40,3),  
(4,'card','paid',100,4),  
(5,'cash','pending',60,5),  
(6,'card','paid',70,6),  
(7,'cash','pending',90,7),  
(8,'card','paid',120,8);
```

insert into reviews values

```
(1,5,'great food',current_timestamp,1,1),  
(2,4,'nice taste',current_timestamp,2,2),  
(3,3,'okay',current_timestamp,3,3),  
(4,5,'excellent',current_timestamp,4,4),  
(5,2,'bad service',current_timestamp,5,5),  
(6,4,'good',current_timestamp,6,6),  
(7,5,'amazing',current_timestamp,7,7),  
(8,4,'nice',current_timestamp,8,8);
```

DDL, DML, DQL AND TRIGGER QUERIES

```
delete from reviews where rating = 2;
```

```
alter table customers add age int;
```

```
alter table customers modify phone_number varchar(15);
```

```
alter table customers drop column age;
```

```
rename table reviews to customer_reviews;
```

```
update restaurants set rating = 5 where id = 5;
```

```
select orders.id , customers.full_name , orders.total_amount from orders  
join customers on orders.customer_id = customers.id;
```

```
select restaurants_name , rating from restaurants where rating > 4;
```

```
select * from orders order by total_amount desc;
```

```
select restaurant_id , count(*) as total_orders from orders group by  
restaurant_id;
```

```
select * from restaurants where restaurants_name like 's%';
```

```
select * from orders where total_amount between 50 and 100;
```

```
select customers.full_name ,restaurants.restaurants_name
,orders.total_amount from orders
join customers on orders.customer_id = customers.id
join restaurants on orders.restaurant_id = restaurants.id;
```

```
select restaurants_name , rating from restaurants where rating = (select
max(rating)from restaurants);
```

```
select customer_id , total_amount from orders where total_amount >
(select avg(total_amount)from orders);
```

```
select customers.full_name ,count(orders.id) as total_orders from
customers
join orders on customers.id = orders.customer_id
group by customers.full_name;
```

```
select * from restaurants where rating = 5;
```

```
select full_name from customers where id in(select customer_id from
orders where order_status = 'delivered');
```

```
delimiter //
```

```
create trigger update_order_status
after update on payments
for each row
begin
    if new.payment_status = 'paid' then
        update orders set order_status = 'delivered' where id = new.order_id;
    end if;
end //
```

```
delimiter ;
```

```
update payments set payment_status = 'paid' where id = 1;
```

```
select * from orders where id = 1;
```